

ATUL KUMAR

✉ Email | 🌐 Portfolio | 🐙 Github | 💼 LinkedIn | 📁 Codechef

EDUCATION

Birla Institute of Technology, Mesra <i>Bachelor of Technology in Chemical Engineering</i>	2023 – Present Ranchi, India
Chinmaya Vidyalaya, Bokaro Steel City <i>Higher Secondary: Class: 11,12</i>	2021 – 2022 Bokaro, Jharkhand

EXPERIENCE

Web Development Intern <i>Zaalima Development (Remote)</i>	Apr 2026 – Present
- Developing a dashboard UI to manage and track 100+ AI-generated extensions, improving workflow visibility and user experience. - Implementing an “Edit Request” feature enabling real-time modification of generated code via prompts. - Building subscription-based feature gating for API-driven functionalities and actively deploying the subscription workflow. - Integrating code sanitization pipelines and conducting ongoing security audits to prevent malicious outputs in AI-generated code.	
Machine Learning Intern <i>Narnetix AI (Remote)</i>	Dec 2025 – Jan 2026
- Built 3+ LLM-based autonomous agents to automate business workflows, reducing manual effort by 95% manual tasks. - Designed 4+ automation pipelines using n8n, Zapier, and REST APIs to streamline operations. - Developed AI-driven analytics modules for predictive insights and anomaly detection. - Integrated AI solutions with 4+ enterprise tools (CRM/ERP), improving operational efficiency by 3.5×.	

PROJECTS

Automated Email Outreach System <i>Python, Pandas, SMTP</i> 🐙	Feb 2026
- Developed a Python CLI tool to automate email outreach to 100+ HR using structured CSV data. - Integrated secure Gmail SMTP with environment variable authentication to prevent credential exposure. - Built data validation and preprocessing pipeline to handle missing/invalid entries. - Implemented rate-limited batch delivery with error handling, reducing manual effort by 90%.	
Dental Caries Segmentation <i>YOLOv8, PyTorch, Computer Vision</i> 🐙	Jan 2026
- Built an end-to-end instance segmentation pipeline for detecting dental caries from radiographic images. - Trained model on 768×768 images using AdamW optimizer with cosine learning rate scheduling. - Achieved 22.6% mask recall using Precision, Recall, and mAP evaluation metrics.	
Market Regime Detection <i>Python, Pandas, NumPy, Scikit-learn</i> 🐙	Dec 2025
- Analyzed NIFTY 50 data (2018–2025) using rolling returns and volatility features. - Applied unsupervised clustering to classify market regimes into bull, bear, and sideways states. - Identified return-volatility patterns to improve interpretability of market conditions.	

TECHNICAL SKILLS

Programming: Python, C, C++
Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Model Evaluation
Libraries: Scikit-learn, NumPy, Pandas, PyTorch
Databases & Tools: MySQL, DBMS, Power BI, MS Excel, Tableau
Technologies: Git, Docker, Jupyter Notebook
Web Development: HTML, CSS, JavaScript, WordPress
Cloud: AWS, Google Cloud

CERTIFICATIONS & JOB SIMULATIONS

Deloitte Data Analytics Job Simulation (Forage)
Data Analyst Bootcamp (Udemy)